Theorem 01. *Let Λ be a nonempty indexing set and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets, each of which is a subset of some universal set U. Then*

1. *For each $\beta \in \Lambda$, $\bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta$.*

2. *For each $\beta \in \Lambda$, $A_\beta \subseteq \bigcup_{\alpha \in \Lambda} A_\alpha$.*

3. $\left(\bigcap_{\alpha \in \Lambda} A_\alpha \right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$

4. $\left(\bigcup_{\alpha \in \Lambda} A_\alpha \right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$

Parts (3) and (4) are known as ***De Morgan's Laws****.*

Theorem 02 . *Let Λ be a nonempty indexing set, let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets, and let B be a set. Then*

1. $B \cap \left(\bigcup_{\alpha \in \Lambda} A_\alpha \right) = \bigcup_{\alpha \in \Lambda} (B \cap A_\alpha)$, *and* Ex.

2. $B \cup \left(\bigcap_{\alpha \in \Lambda} A_\alpha \right) = \bigcap_{\alpha \in \Lambda} (B \cup A_\alpha)$.

End of Dis

Pairwise Disjoint Families of Sets

Two sets A and B to be disjoint provided that $A \cap B = \emptyset$.
In a similar manner, if Λ is a nonempty indexing set and $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ is an indexed family of sets, we can say that this indexed family of sets is **disjoint** provided that $\bigcap_{\alpha \in \Lambda} A_\alpha = \emptyset$. However, we can use the concept of two disjoint sets to define a somewhat more interesting type of "disjointness" for an indexed family of sets.

Definition. Let Λ be a nonempty indexing set, and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets. We say that $\mathcal{A}$ is **pairwise disjoint** provided that for all α and β in Λ, if $A_\alpha \neq A_\beta$, then $A_\alpha \cap A_\beta = \emptyset$.

Eg Let $A_n = \{x \in \mathbb{R} \mid n < x < n+1\}$, $n \in \mathbb{N}$

"$\mathcal{A}$ is pairwise disjoint" $\underset{\text{Defn}}{\Longleftrightarrow}$ "$\forall \alpha, \beta \in \Lambda$, $A_\alpha \neq A_\beta \Rightarrow \underbrace{A_\alpha \cap A_\beta = \emptyset}$" $A_1 \cap A_2 = \emptyset$ $A_i \cap A_j = \emptyset$.

Note:

The **Cartesian product** (or just **product**) of two sets A and B is the set

$$A \times B := \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

The product of a finite family $A_1, A_2, \ldots, A_n$ of sets is the set

$$A_1 \times A_2 \times \cdots \times A_n := \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}.$$

The product of a finite family $A_1, A_2, \ldots, A_n$ of sets is the set

$$\prod_{i=1}^{n} A_i = A_1 \times A_2 \times \cdots \times A_n$$
$$= : \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}.$$

1) Proof of " $\forall \beta \in \Lambda, \bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta$ "

Let $\beta \in \Lambda$ and x be an arbitrary element in U.

Suppose $x \in \bigcap_{\alpha \in \Lambda} A_\alpha$.

Then $x \in A_\alpha$ for all $\alpha \in \Lambda$

$\therefore x \in A_\beta$ (Since, $\beta \in \Lambda$)

$\therefore x \in \bigcap_{\alpha \in \Lambda} A_\alpha \Rightarrow x \in A_\beta$

$\therefore \bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta$

$\therefore \forall \beta \in \Lambda, \bigcap_{\alpha \in \Lambda} A_\alpha \subseteq A_\beta$ ☒

3) " Proof of $\left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$ "

Let $x \in U$ be an arbitrary element

$$
\begin{aligned}
x \in \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c &\iff x \in U \wedge x \notin \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right) \\
&\iff x \in U \wedge \left(\sim \left(x \in \bigcap_{\alpha \in \Lambda} A_\alpha\right)\right) \\
&\iff x \in U \wedge \left(\sim \left(\forall \alpha \in \Lambda, x \in A_\alpha\right)\right) \\
&\iff x \in U \wedge \left(\exists \alpha_0 \in \Lambda, x \notin A_{\alpha_0}\right) \\
&\iff \exists \alpha_0 \in \Lambda, x \in U \wedge x \notin A_{\alpha_0} \\
&\iff \exists \alpha_0 \in \Lambda, x \in A_{\alpha_0}^c \\
&\iff x \in \bigcup_{\alpha \in \Lambda} A_\alpha^c
\end{aligned}
$$

Hence, $\left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$ ☒

Proof of Theorem 2.1. $B \cap \left(\bigcup_{\alpha \in \Lambda} A_\alpha\right) = \bigcup_{\alpha \in \Lambda} (B \cap A_\alpha)$

Proof. Assume that E is the Universal set.

Let x be an arbitrary element in E.

$$
\begin{aligned}
x \in B \cap \left(\bigcup_{\alpha \in \Lambda} A_\alpha\right) &\iff x \in B \wedge x \in \left(\bigcup_{\alpha \in \Lambda} A_\alpha\right) \\
&\iff x \in B \wedge \left(\exists \alpha_0 \in \Lambda, x \in A_{\alpha_0}\right) \\
&\iff \exists \alpha_0 \in \Lambda, (x \in B \wedge x \in A_{\alpha_0}) \\
&\iff \exists \alpha_0 \in \Lambda, x \in (B \cap A_{\alpha_0}) \\
&\iff x \in \bigcup_{\alpha \in \Lambda} (B \cap A_\alpha)
\end{aligned}
$$
☒

Ex. Prove that $\mathbb{N} = \bigcup_{n=1}^{\infty} \{n\}$.

01. For each natural number n, let $A_n = \{x \in \mathbb{R} \mid n-1 < x < n\}$. Prove that $\{A_n \mid n \in \mathbb{N}\}$ is a pairwise disjoint family of sets and that $\bigcup_{n \in \mathbb{N}} A_n = (\mathbb{R}^+ - \mathbb{N})$.

02. For each natural number n, let $A_n = \{k \in \mathbb{N} \mid k \geq n\}$. Determine if the following statements are true or false. Justify each conclusion.

(a) For all $j, k \in \mathbb{N}$, if $j \neq k$, then $A_j \cap A_k \neq \emptyset$.

(b) $\bigcap_{k \in \mathbb{N}} A_k = \emptyset$

03. Give an example of an indexed family of sets $\{A_n \mid n \in \mathbb{N}\}$ such all three of the following conditions are true:

(i) For each $m \in \mathbb{N}$, $A_m \subseteq (0, 1)$;

(ii) For each $j, k \in \mathbb{N}$, if $j \neq k$, then $A_j \cap A_k \neq \emptyset$; and

(iii) $\bigcap_{k \in \mathbb{N}} A_k = \emptyset$.

04. Prove the following:

(a) Let $\{A_i : i \in I\}$ and $\{B_i : i \in I\}$ be indexed families of sets with the same indexed set I. Suppose $A_i \subseteq B_i$ for all $i \in I$.

(i) $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} B_i$.

(ii) $\bigcap_{i \in I} A_i \subseteq \bigcap_{i \in I} B_i$.

(b) Let $\{A_i : i \in I\}$ and $\{B_j : j \in J\}$ be indexed families of sets. If there is an $i_0 \in I$ such that $A_{i_0} \subseteq B_j$ for all $j \in J$, then $\bigcap_{i \in I} A_i \subseteq \bigcap_{j \in J} B_j$.

(c) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets. Then $A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i)$.

(d) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets. Then $A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i)$.

(f) Suppose that A is a set and that $\{B_i : i \in I\}$ is an indexed family of sets. Then $A \setminus \bigcap_{i \in I} B_i = \bigcup_{i \in I} (A \setminus B_i)$.

Eg: Let $n \in \mathbb{N}$ and $A_n = \{x \mid x \in \mathbb{R},\ n < x < n+1\}$.
Prove that $\forall i, j \in \mathbb{N}$ if $i \neq j$, then $A_i \cap A_j = \emptyset$.

Proof: Suppose "$\underbrace{\forall i, j \in \mathbb{N} \text{ if } i \neq j,}$ then $A_i \cap A_j = \emptyset$" is false

Then, $\exists i_0, j_0 \in \mathbb{N}$ and $i_0 \neq j_0$, $A_{i_0} \cap A_{j_0} \neq \emptyset$

Since $A_{i_0} \cap A_{j_0} \neq \phi$, $\exists x_0 \in A_{i_0} \cap A_{j_0}$

$\therefore$ $x_0 \in A_{i_0}$ and $x_0 \in A_{j_0}$

$\therefore$ $i_0 < x_0 < i_0+1$ and $j_0 < x_0 < j_0+1$

Since $i_0 \neq j_0$, $i_0 > j_0$ or $i_0 < j_0$

Case I $i_0 > j_0$: —— $i_0 - j_0 \geq 1$ —(*)

$j_0 < x_0 < j_0+1 \leq i_0 < x_0$

$\therefore$ $x_0 < x_0$ is a contradiction.

$\therefore$ $\forall i, j \in \mathbb{N}$ if $i \neq j$, then $A_i \cap A_j = \phi$

Case II $i_0 < j_0$: $j_0 - i_0 \geq 1$

$\underbrace{x_0 > j_0 \geq 1 + i_0 > x_0}$

4 Relations

A relation is an association between objects. A book on a table is an example of the relation of one object being *on* another. It is especially common to speak of relations among people. For example, one person could be the uncle of another. In mathematics, there are many relations such as equals and less-than that describe associations between numbers. To formalize this idea, we make the next definition.

Definition

Let A and B are two non empty sets. A relation $\mathcal{R}$ from A to B is a subset of $A \times B$.

- In other words, a relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B.
- If $(a, b) \in \mathcal{R}$, then we say that a is related to b by $\mathcal{R}$.
- $a\mathcal{R}b$ write to express that $(a, b) \in \mathcal{R}$ and $a \not\mathcal{R} b$ to express that $(a, b) \notin \mathcal{R}$.
- A relation on a set A is a subset of $A \times A$.

Example :

Let A be the set $\{1, 2, 3, 4\}$ and the relation $\mathcal{R}$ defined on a set A by $\mathcal{R} = \{(a, b) \mid a \text{ devides } b\}$.
Thus,

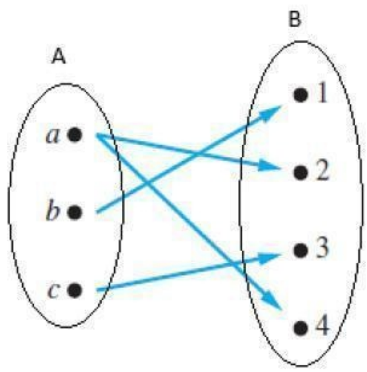
$$\mathcal{R} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$$

Representing relations : Relations can be represented graphically, using arrows to represent ordered pairs. In other words, to connecte the domain and codomain elements. A relation can relate an element with more than one value.

Let $A = \{a, b, c,\}$, $B = \{1, 2, 3, 4\}$ and the relation defined as

$$\mathcal{R} = \{(a,2), (a,4), (b,1), (c,3)\} \subseteq A \times B$$

Note that R is a relation from A to B.



Note:

A set R is an (**n-ary**) **relation** if there exist sets $A_1, A_2, \ldots, A_n$ such that

$$R \subseteq A_1 \times A_2 \times \cdots \times A_n.$$

In particular, R is a **unary** relation if $n = 1$ and a **binary** relation if $n = 2$.
If $R \subseteq A \times A$ for some set A, then R is a **relation on** A.

Definition: Let $R \subseteq A \times B$.

The **domain** of R is the set

$$\mathrm{dom}(R) = \{x \in A : \exists y \text{ such that } y \in B \wedge (x, y) \in R\},$$

and the **range** of R is the set

$$\mathrm{ran}(R) = \{y \in B : \exists x \text{ such that } x \in A \wedge (x, y) \in R\}.$$

Example:

Let $A = \{1, 2, 3, 4, 5, 6\}$, and define R by xRy iff $x \lneq y$ and x divides y;

$$R = \{(1, 2), (1, 3), \ldots, (1, 6), (2, 4), (2, 6), (3, 6)\},$$

and

$$\mathrm{dom}\, R = \{1, 2, 3\}, \ \mathrm{ran}\, R = \{2, 3, 4, 5, 6\}.$$

Note:

Let A and B be two sets. Suppose ρ is a relation from A to B (i.e. ρ is a subset of $A \times B$). Then, ρ is a set of ordered pairs where each first element comes from A and each second element from B.

Thus, we denote it with an ordered pair (a, b), where $a \in A$ and $b \in B$.

We also denote the relationship with $a\rho b$, which is read as a related to b.

The ***domain of ρ*** is the set of all first elements in the ordered pair and the ***range of ρ*** is the set of all second elements in the ordered pair.

Notation:

Relation are usually denoted by $\boldsymbol{R}$ or the Greek letters ρ, σ, τ ...etc

Example 1.

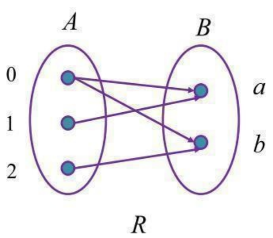
A : the set of students in the Faculty.

B : the set of index numbers.

$R = \{\, (a, b) : a \in A,\ b \in B,\ a \text{ is enrolled in Index no. } b \,\}$

Example 2. Let $A=\{0, 1, 2\}$ and $B=\{a, b\}$, then $\{(0,a),(0,b),(1,a),(2,b)\}$ is a relation R from A to B.

This means, for instance, that $0Ra$, but that $1\not R b$.



$$R \subseteq A \times B = \{\, (0,a)\, ,\, (0,b)\, ,\, (1,a)\, ,\, \underbrace{(1,b)}_{\notin R}\, ,\, \underbrace{(2,a)}_{\notin R}\, ,\, (2,b) \}$$

dom(R) = {0, 1, 2}

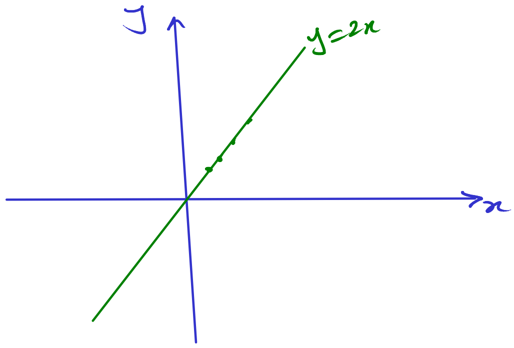
ran(R) = {a, b}

Example 4.: Let $A = \{1, 2, 3, 4\}$ and ρ be defined as:
$\rho = \{(1, 2), (2, 3), (2, 1), (3, 2), (3, 3)\}$. Then ρ is a relation on A.

Example 5 :
Let $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$.
Let $\rho = \{(1, x), (2, x), (3, y), (3, z)\}$.
Then ρ is a relation from A to B.
The domain of ρ is: $\{1,2,3\}$
The range of ρ is : $\{x, y, z\}$

Example 6 :
Suppose we say that two countries are ***adjacent*** if they have some part of their boundaries common. Then, "***is adjacent to***", is a relation ρ on the countries on the earth. Thus, we have,
(India, Nepal) $\in \rho$, but
(Japan, Sri Lanka) $\notin \rho$.

Example 7: Let $A = \mathbb{R}$ and ρ be defined as:
$\rho = \{(x, y) \mid x, y \in \mathbb{R}$ and $y = 2x\}$. Then ρ is a relation on A.
$= \{(x, 2x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$



Example 8 : Let $A = \mathbb{R}$ and ρ be defined as:
$\rho = \{(x, y) \mid x, y \in \mathbb{R}$ and $x^2 + y^2 \leq 1\}$. Then ρ is a relation on A.

Example 9 : Let $A = \mathbb{R}$ and ρ be defined as:
$\rho = \{(x, y) \mid x, y \in \mathbb{R}$ and $y < x + 1\}$. Then ρ is a relation on A.

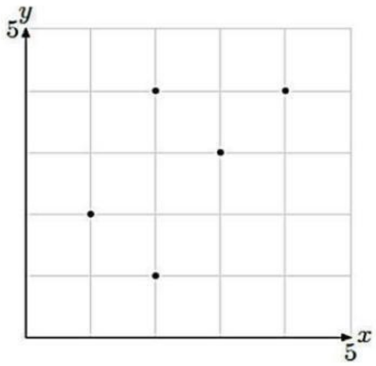
Example 10 : Let $A = \mathbb{R}$ and ρ be defined as:
$\rho = \{(x, y) \mid x, y \in \mathbb{R}$ and $y^2 \geq x\}$. Then ρ is a relation on A.

End of D5

Graph of the relation

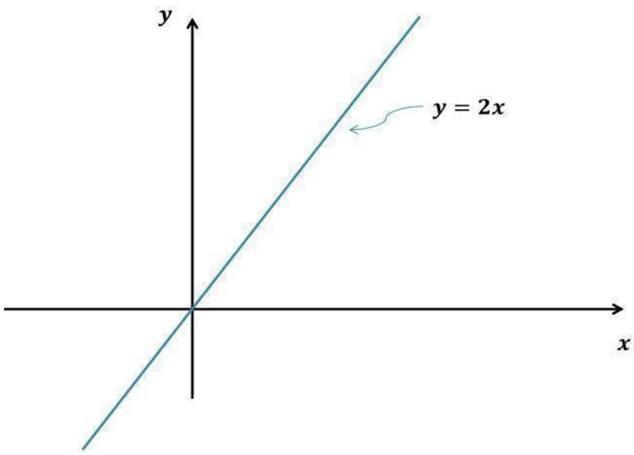
Let $\rho = \{(1, 2), (2, 1), (2, 4), (3, 3), (4, 4)\}$

Consider the graph of the relation ρ

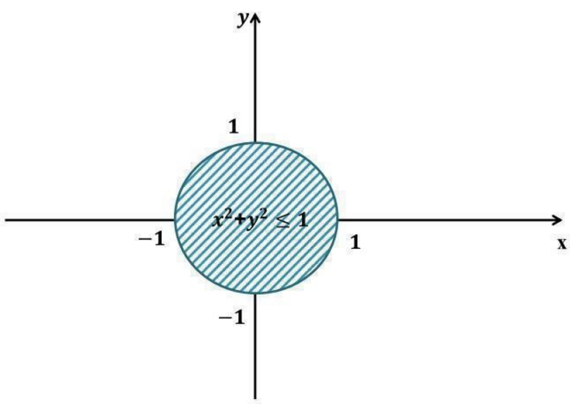


ρ has Domain $= \{1, 2, 3, 4\}$ and Range $= \{1, 2, 3, 4\}$.

$\rho = \{(x,y) \mid x, y \in \mathbb{R} \text{ and } y = 2x\}.$



$\rho = \{(x,y) \mid x, y \in \mathbb{R} \text{ and } x^2+y^2 \leq 1\}.$

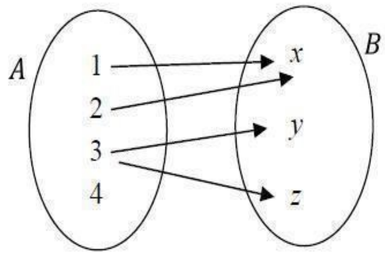


Representation of relations:

Consider the relation in Example 2

Let $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$.
Let $\rho = \{(1,x), (2,x), (3,y), (3,z)\}$.

This relation can be represented pictorially as well, as follows:

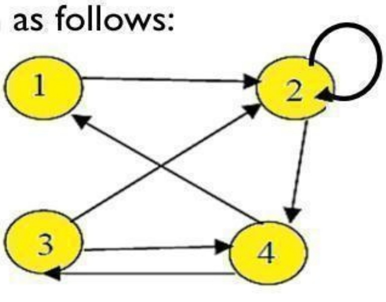


If the relation is from a finite set to itself, there is another way of pictorial representation, known as **digraph**

Definition: A directed graph (digraph) consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs).

For example, let A = {1, 2, 3, 4} and σ be a relation from A to itself, defined as follows:
$\sigma = \{(1, 2), (2, 2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\}$
Then, the diagraph of σ is drawn as follows:

vertex : 1, 2, 3, 4
edge : (1,2), (2,2), (2,4),
(3,2), (3,4), (4,1), (4,3)



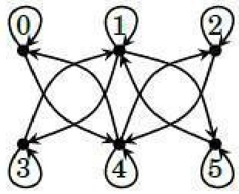
Exercises

1. Let $A = \{0,1,2,3,4,5\}$. Write out the relation R that expresses $>$ on A. Then illustrate it with a diagram.

2. Let $A = \{1,2,3,4,5,6\}$. Write out the relation R that expresses $\mid$ (divides) on A. Then illustrate it with a diagram.

3. Let $A = \{0,1,2,3,4,5\}$. Write out the relation R that expresses $\geq$ on A. Then illustrate it with a diagram.

4. Here is a diagram for a relation R on a set A. Write the sets A and R.



5. Here is a diagram for a relation R on a set A. Write the sets A and R.